American Computer Science League

2021-2022 ● Contest 1: Short Problems ● Classroom Division

1. Computer Number Systems

Convert 2122_{10} to octal.

A. 4211
B. 4121
C. 1224
D. 4112
E. 2114

2. Computer Number Systems

The Fibonacci sequence {0, 1, 1, 2, 3, 5, 8, 13, …} is formed by adding the two previous terms: 5 + 8 = 13. Which number in the sequence when converted to binary would include the 4th occurrence of the substring "10" when considering each number separately?

A. 8
B. 13
C. 21
D. 34
E. 55

3. Recursive Functions

Find $f(10)$, given

$$f(x) = \begin{cases} f(x-2) + 3 & \text{if } x \geq 7 \\ f(x-1) - 1 & \text{if } 3 < x < 7 \\ x^2 & \text{if } x \leq 3 \end{cases}$$

A. 6
B. 7
C. 8
D. 9
E. 12

4. Recursive Functions

Find $f(20, 12)$, given

$$f(x,y) = \begin{cases} f(x-3,\ y+1) + 3 & \text{if } x > y \\ 2 * f(x+2,\ y-3) - 1 & \text{if } x = y \\ x + y & \text{if } x < y \end{cases}$$

A. 29
B. 34
C. 63
D. 23
E. 60

5. What Does This Program Do? (Branching)

What is the output when this program is executed?

```
a = 0 : b = 2 : c = 4 : d = 6
if c > d / b then
   c = b * d
end if
if b ^ 2 < c / d then
  c = b ^ 2
else
   b = b ^ 2
end if
if c - d < a * b then
   a = b + c
end if
if a ^ 2 > 1 then
   b = b + a
else
   b = b + d
end if
output c / d - b + a ^ 3 * d
```

A. -8
B. 0
C. -2
D. -4
E. 2

6. Computer Number Systems

Convert 2122_{10} to hexadecimal.

A. A48
B. B37
C. 849
D. 84A
E. 73A

7. Computer Number Systems

The Fibonacci sequence {0, 1, 1, 2, 3, 5, 8, 13, …} is formed by adding the two previous terms: 5 + 8 = 13. Which number in the sequence when converted to octal would have the 6th occurrence of the digit "1" when counting every digit?

A. 34
B. 55
C. 89
D. 144
E. 233

8. Recursive Functions

Find $f(20)$, given

$$f(x) = \begin{cases} 2 * f(x-3) + 1 & \text{if } x \geq 15 \\ \left[\frac{f(x+2)}{3}\right] & \text{if } 12 < x < 15 \\ 2 * x - 3 & \text{if } x \leq 12 \end{cases}$$

where [y] returns the greatest integer less than or equal to y

A. 43
B. 39
C. 19
D. 14
E. 9

9. Recursive Functions

Find $f(f(f(f(20))))$, given

$$f(x) = \begin{cases} f\left(\frac{x-2}{2}\right) - 1 & \text{if } x \text{ is even} \\ 2 * x & \text{if } x \text{ is odd} \end{cases}$$

A. 34
B. 18
C. 17
D. 12
E. 9

10. What DoesThis Program Do? (Branching)

What is printed when this program is executed?

```
a = 10 : b = 4 : c = 0 : d = 6
if a > b + d then
   b = b + d
else
   d = a - d
end if
if c ^ 2 < c ^ 4 then
   d = d * 2
else
   c = 1
end if
if b / d == c then
   c = c * 2
else
   a = a / c
end if
if b > c && d < b then
   d = d / 4
else
   a = c ^ 2 / d
end if
if d - b < a + b || b + c == d then
   d = d / 2
end if
x = a - b * c + d
output x
```

A. 7
B. 3
C. 1
D. -5
E. -9